

Diagnostic information - air conditioning

Temperature sensor **power supply**

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, clear the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Short circuit to B+
- Short circuit to ground
- Open circuit

Fault effects

- ♦ Restricted function of **air conditioner**

See document "Tequipment Installation and Conversion Instructions, 7201 Sports bucket seat (I-no. 388/389)".

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Diagnosis information - Air-conditioning system

Power supply Sun sensor

Diagnostic conditions

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In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on
- Power supply **Air-conditioning system regulator** 9... 16 V

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines

Fault setting condition(s)

Power supply:

- Short circuit to B+ > 4 s
- Short circuit to ground > 4 s
- Open circuit > 4 s

Fault effects

- ♦ **Air-conditioning system** Limited function

Diagnosis information - Air-conditioning system

Power supply Sun sensor

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The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

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Power supply:

- Short circuit to B+ > 4 s
- Short circuit to ground > 4 s
- Open circuit > 4 s

Fault effects

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Power supply:

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- Power supply **Air-conditioning system regulator** 9... 16 V

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines

Fault setting condition(s)

Power supply:

- Short circuit to B+ > 4 s
- Short circuit to ground > 4 s
- Open circuit > 4 s

Fault effects

- ♦ **Air-conditioning system** Limited function

Diagnosis information - Air-conditioning system

Power supply **Pressure sensor**

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Short circuit to B+
- Short circuit to ground
- Open circuit

Fault effects

- ♦ **Air-conditioning system** Limited function

Diagnosis information - Air-conditioning system

Power supply **Pressure sensor**

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Short circuit to B+
- Short circuit to ground
- Open circuit

Fault effects

- ♦ **Air-conditioning system** Limited function

Diagnosis information - Air-conditioning system

Pressure sensor Frequency (f)

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Frequency (f), incorrect

Fault effects

- ♦ **Air-conditioning system** Limited function

Diagnosis information - Air-conditioning system

Central locking system button Short circuit to ground

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Short circuit to ground
- **Button** pressed > 16 Second(s)

Fault effects

- ♦ **Central locking system button** No function

Diagnosis information - Air-conditioning system

Adjusting motor blocked

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- blocked

Fault effects

- ♦ **Adjusting motor** No function

Diagnosis information - Air-conditioning system

Adjusting motor Faulty

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Faulty
- Current permanent Too high

Fault effects

- ♦ **Adjusting motor** No function

Diagnosis information - Air-conditioning system

Adjusting motor Measurement run

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- **Adjusting motor** Step

too high / too low

Fault effects

- ♦ **Adjusting motor** No function

Diagnosis information - Air-conditioning system

Adjusting motor blocked

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- blocked

Fault effects

- ♦ **Adjusting motor** No function

Diagnosis information - Air-conditioning system

Servo motor Flap control box

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on
- Power supply checked
- Internal function test **Servo motor**

Fault setting condition(s)

Servo motor Mechanical fault

- blocked
- Fault present for 1 ... 11 s

Servo motor Electrical fault

- Fault present for 1 ... 11 s

Servo motor not programmed

- Fault present for 1 ... 11 s

Servo motor Message missing

- Fault present for 1 ... 20 s

Possible fault causes

Servo motor Mechanical fault

- Flap or Engine blocked
- Axle or limit stop broken

Servo motor Electrical fault

- Short circuit to B+/ground or open circuit in the line

Servo motor not programmed

- Component Faulty
- no/incorrect coding/calibration

Servo motor Message missing

- Component Faulty
- Open circuit LIN line

Fault effects

- ◆ Limited function
- ◆ Function **Servo motor** Faulty

Diagnosis information - Air-conditioning system

Adjusting motor Measurement run

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- **Adjusting motor** Step

too high / too low

Fault effects

- ♦ **Adjusting motor** No function

Diagnosis information - Air-conditioning system

Adjusting motor blocked

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

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Fault setting condition(s)

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Fault effects

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Adjusting motor Faulty

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- Ignition on

Fault setting condition(s)

- Faulty
- Current permanent Too high

Fault effects

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Adjusting motor Measurement run

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Fault setting condition(s)

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